



RV College of Engineering®
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Bengaluru - 560059, Karnataka, India

Question Paper ID: 718383

Course Code: CS374TFA

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RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

VII Semester B.E. Regular Examination February/March 2026

COMPUTER SCIENCE AND ENGINEERING

Deep Learning

Time : 3 Hours

Instructions to the students

Maximum Marks : 100

1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

Question No	Question	M	CO	BT
1.1	Explain why a perceptron cannot represent XOR?	02	1	2
1.2	Differentiate between stochastic gradient descent and standard gradient descent.	02	1	4
1.3	Compare tiled convolution with locally connected layers in terms of parameter sharing and memory requirements.	02	2	4
1.4	Analyze how increasing filter depth changes model capability.	02	1	4
1.5	What is teacher forcing in recurrent neural networks? Why is it used during training?	02	1	2
1.6	Explain one way in which recurrent units can be forced to operate at different time scales by modifying network connectivity.	02	3	2
1.7	Explain why does depth improve the representational power of autoencoders compared to shallow architectures?	02	3	2
1.8	Explain the connection between denoising autoencoders (DAE) and contractive autoencoders (CAE).	02	2	2
1.9	Why is LeNet considered an important milestone in the development of deep learning?	02	1	2
1.10	Apply DenseNet's dense connectivity concept to explain how it prevents the vanishing-gradient problem in deep networks	02	4	3

Part B

Question No	Question	M	CO	BT
	A neuron has three inputs: $x_1 = 0.5, x_2 = -1.0, x_3 = 2.0$ with the corresponding weights $w_1 = 0.8, w_2 = -0.4, w_3 = 0.3$, and bias $b = 0.2$. Compute the output of the neuron for the following activation functions:			
2a	i) Threshold (Heaviside) function ii) Sigmoid function iii) Hyperbolic Tangent Function Show all intermediate steps, including the calculation of the induced local field, and final outputs.	10	1	3
2b	Explain gradient-based optimization flow beginning from forward propagation to weight update. Compare GD vs SGD and argue when each is preferable.	06	2	2
3a	Explain pooling in CNNs. Describe types of pooling, why pooling is beneficial, and its role in translation invariance.	10	2	2
3b	A CNN trained on handwritten digits fails to identify digits that are slightly rotated. Explain why and suggest fixes.	06	2	3
OR				
4a	Define parameter sharing and explain how CNNs exploit it. Consider a CNN that processes a $64 \times 64 \times 3$ image:	06	2	2
4b	Layer 1: Conv, 5×5 kernel, 8 filters, stride=1, padding='same' Layer 2: Max pool 2×2 stride=2 Layer 3: Flatten Layer 4: Fully connected with 10 outputs Calculate output size after each stage and justify.	10	2	3
5a	A machine translation system is required to translate English sentences into German sentences, where the input and output sequences differ significantly in length. A practitioner initially uses a standard RNN that maps sequences of equal length but observes poor translation quality. (i) Explain how an encoder-decoder (sequence-to-sequence) RNN architecture addresses this problem. (ii) Discuss a key limitation of this architecture when translating long sentences and describe how attention mechanisms overcome this limitation.	06	3	4
5b	Describe and compare the major design patterns of recurrent neural networks.	10	3	2
OR				

6a	A company's shallow RNN model fails to capture long-term dependencies in complex sequential data. How can deep recurrent network architectures improve performance in this situation? Discuss with reference to deeper transformations, hierarchical hidden layers, and skip connections.	10	4	4
6b	Which components of the LSTM architecture are essential for controlling time scale and forgetting? Explain how Gated Recurrent Units (GRUs) provide an alternative architecture to achieve dynamic control of memory and forgetting, including the governing equations.	06	3	4
7a	Explain how depth and layer size affect the representational power of autoencoders. Why do deep autoencoders generally outperform shallow autoencoders in learning useful representations?	06	4	2
7b	Describe Regularized Autoencoders and explain how sparsity, denoising, and Contractive penalties help autoencoders learn meaningful representations.	10	3	2
OR				
8a	Describe stochastic autoencoders and explain their relationship with latent variable models. Why are encoder and decoder distributions not always compatible with a unique joint distribution?	06	3	2
8b	A classification model trained on high-dimensional data suffers from overfitting and high computational cost. The researcher introduces an autoencoder as a preprocessing step before classification.	04	4	4
8c	Examine, how using an autoencoder can improve both classification performance and computational efficiency. Compare sparse autoencoders and denoising autoencoders with respect to their training objectives and the type of representations they learn.	06	3	4
9a	Explain the architecture of Residual Networks (ResNet) and discuss how skip connections improve training of deep neural networks?	10	2	2
9b	Discuss the advantages of hyperparameter tuning with respect to model performance, overfitting control, and resource utilization.	06	2	2
OR				
10a	Explain how dataset augmentation helps improve the generalization ability of machine learning models. Illustrate your answer with suitable examples.	06	2	2
10b	With a neat diagram, explain the architecture and key design principles of VGGNet.	10	2	2